

VIMAL RAM K

Senior Machine Learning Engineer

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PROFESSIONAL SUMMARY

Senior Machine Learning Engineer with 5 years of enterprise experience designing, deploying, and troubleshooting production ML systems. Skilled in Python, TensorFlow, PyTorch, Hugging Face transformers, NLP, Computer Vision, and Generative AI. Proven track record in building scalable inference APIs with Docker and FastAPI, implementing MLOps practices (MLflow, CI/CD, monitoring), and resolving production issues through root-cause analysis. Strong cloud familiarity with Google Cloud (Vertex AI), AWS, and container orchestration using Kubernetes.

TECHNICAL SKILLS

Languages & Core	Python
Frameworks & Libraries	TensorFlow, PyTorch, Keras, Hugging Face Transformers, Xgboost
AI Domain Expertise	Machine Learning Algorithms, Deep Learning, Natural Language Processing (NLP), Computer Vision, Generative AI, CNN, Rnn
MLOps & Infrastructure	FastAPI, Docker, Kubernetes, MLflow, CI/CD, Cloud Storage

WORK EXPERIENCE

Senior Machine Learning Engineer

Feb 2025 – Present

IBM

Bengaluru, India

- Designed and deployed ultra-low-latency large language model (LLM) ecosystems and enterprise-grade generative AI pipelines using PyTorch and TensorFlow.
- Built and optimized a distributed Model Hosting Platform to manage 100+ high-parameter models achieving 99.9% uptime.
- Developed model compression workflows and performance optimizations reducing inference latency by 35% and GPU memory usage by 25%.
- Implemented end-to-end, zero-downtime ML pipelines from data ingestion to production inference, integrating CI/CD and MLflow for experiment tracking.
- Established automated drift detection, shadow deployments, and A/B testing to preserve model integrity and reliability.
- Authored internal debugging guides and collaborated cross-functionally to resolve production issues and improve observability.

Machine Learning Engineer

Feb 2022 – Feb 2025

Capgemini

Bengaluru, India

- Developed end-to-end AI solutions and ML-integrated web applications with React frontend and FastAPI/Flask backends.
- Performed data preprocessing, feature engineering, and model experimentation for classification and regression tasks; diagnosed and resolved model accuracy degradation.
- Integrated ML inference services into production via REST APIs and reproduced customer-reported integration failures in service-oriented architectures.
- Deployed scalable ML applications on cloud platforms (Azure, AWS) and implemented caching and load-balancing to support high request volumes.
- Built reusable ML pipelines, automation scripts, and diagnostic tooling improving development velocity and support turnaround.

FEATURED PROJECTS

Enterprise RAG Engine & Low-Latency LLM Infrastructure

Production Systems / GitHub

- Architected a high-throughput Retrieval-Augmented Generation (RAG) system to automate processing of over 10,000+ enterprise PDFs.
- Engineered an asynchronous tf.data ingestion pipeline to remove CPU bottlenecks during document chunking and tokenization.
- Designed a low-latency vector search layer using custom tf.linalg operations and Cosine Similarity for precise context retrieval.
- Integrated with a high-performance vLLM engine and managed KV-cache allocation to ensure 99.9% uptime under high-concurrency loads.

Autonomous Multi-Agent AI Workflow Engine

LangGraph Orchestration / GitHub

- Developed a stateful multi-agent orchestration framework using LangGraph and FastAPI to execute multi-step analytical workflows.
- Integrated NeMo Guardrails and OpenTelemetry tracing to enforce PII masking, prevent hallucination, and achieve 100% auditability across LLM calls.
- Reduced complex task execution time by 40% and eliminated out-of-scope model responses in production simulations.

- Ultra-Low-Latency Multimodal LLM Inference Service

Distributed Systems / GitHub

 - Deployed open-weights LLMs using vLLM continuous batching and AWQ 4-bit quantization on Ray/Kubernetes clusters.
 - Configured paged attention and prefix caching, reducing Time-To-First-Token (TTFT) to sub-50ms under heavy concurrent user loads.
 - Improved GPU memory utilization by 3.5x and cut inference hosting costs by 50% compared to standard FP16 deployment baselines.

MLOPS & INFRASTRUCTURE HIGHLIGHTS

- **Model Observability & Monitoring:** Implemented automated metrics tracking for model drift, latency metrics, throughput bottlenecks, and accuracy degradation using MLflow and OpenTelemetry.
- **Scalable Deployment Architecture:** Specialized in containerized deployment via Docker and Kubernetes, building auto-scaling RESTful inference services backed by caching layers and load balancing.
- **Continuous Integration / CI/CD:** Engineered zero-downtime automated deployment pipelines for model releases, feature store management, and regression testing.

CERTIFICATIONS & COMPETENCIES

Deep Learning & GenAI	Specialized in Transformer Architectures, LLM Fine-Tuning, and RAG Implementations
Cloud & Orchestration	Production Workflows across Google Cloud Platform (Vertex AI), AWS, and Docker Containers
Software Standards	Agile / Scrum Methodology, Automated Testing, Root-Cause Debugging, and Code Reviews

EDUCATION & KEY ACHIEVEMENTS

- B.E. in Computer Science and Engineering (CSE)

Nandha College of Technology, Erode, Tamil Nadu

2016 – 2020

Marks: 72%
- **Generative AI Impact:** Improved semantic accuracy of generative AI pipelines by 40% across enterprise workflows.
 - **Infrastructure Reliability:** Maintained high-parameter model hosting infrastructure supporting 100+ production models with 99.9% uptime.
 - **Efficiency Gain:** Reduced GPU memory footprint by 3.5x and cut inference hosting costs by 50% using AWQ 4-bit quantization.